Emerging Topics in Machine Learning and Artificial Intelligence

Seminar Introduction and Topic Overview

Prof. Dr. Dieter Kranzlmüller PD Dr. Andre Luckow

MNM-Team, LMU München

Winter Semester 2026/27

1

Organization

Objectives, rules and grading criteria

2

Motivation

Algorithms, data and compute as drivers of AI progress

3

16 Seminar Topics

Research question, key concepts and three core papers for every topic

4

Topic Selection

Mentors, preference submission and how to work with your topic

Organization

Objectives of the Seminar and Course Material

A PRACTICAL INTRODUCTION TO SCIENTIFIC WORK

1. Literature research: library and publication databases (Google Scholar, ResearchGate, arXiv, AI)
2. Rating and classification of publications
3. Understanding methodologies
4. Relating the topic to HPC and AI research at LMU and LRZ
5. Reviewing scientific papers

COURSE MATERIAL (HAUPTSEMINAR AND PROSEMINAR)

Moodle:
moodle.lmu.de/course/view.php?id=47259

Enrollment key:
LMU#ETMLAI#2026

Organization

A Seminar Organized Like a Conference

WORKFLOW

STEP 1
Write
Prepare a conference paper (technical report) on your topic

STEP 2
Submit
Submit the paper to the program committee

STEP 3
Review
Peer review the papers of other participants

STEP 4
Revise
Receive reviews of your paper and prepare the camera-ready version

STEP 5
Present
Block seminar on **January 29/30, 2027**

TWO TRACKS

Proseminar (Bachelor)

- Work in groups of two students
- Groups are assigned by the system

Hauptseminar (Master)

- Each student submits a single-author paper

Organization

Let's Have a Conference: Example Call for Papers

IEEE Big Data 2022 Call for Papers

2022 IEEE International Conference on Big Data (IEEE BigData 2022)
http://bigdataieee.org/BigData2022/
December 17-20, 2022

IEEE BigData 2022 – Osaka, Japan
(may move to hybrid or online depending on the pandemic situation by then)

In recent years, "Big Data" has become a new ubiquitous term. Big Data is transforming science, engineering, medicine, healthcare, finance, business, and ultimately our society itself. The IEEE Big Data conference series started in 2013 has established itself as the top tier research conference in Big Data.

- The first conference IEEE Big Data 2013 had more than 400 registered participants from 40 countries (http://bigdataieee.org/BigData2013/) and the regular paper acceptance rate is 17.0%.
- The IEEE Big Data 2019 (http://bigdataieee.org/BigData2019/ , regular paper acceptance rate: 18.7%) was held in Los Angeles, CA, Dec 9-12, 2019 with close to 1200 registered participants from 54 countries.
- The IEEE Big Data 2020 (http://bigdataieee.org/BigData2020/ , regular paper acceptance rate: 15.7%) was held online, Dec 10-13, 2020 with close to 1100 registered participants from 50 countries.
- The IEEE Big Data 2021 (http://bigdataieee.org/BigData2021/ , regular paper acceptance rate: 19.9%) was held online, Dec 15-18, 2021 with close to 1089 registered participants from 52 countries

The 2022 IEEE International Conference on Big Data (IEEE BigData 2022) will continue the success of the previous IEEE Big Data conferences. It will provide a leading forum for disseminating the latest results in Big Data Research, Development, and Applications.

We solicit high-quality original research papers (and significant work-in-progress papers) in any aspect of Big Data with emphasis on 5Vs (Volume, Velocity, Variety, Value and Veracity), including the Big Data challenges in scientific and engineering, social, sensor/IoT/IoE, and multimedia (audio, video, image, etc.) big data systems and applications. The conference adopts single-blind review policy. We expect to have a very high quality and exciting technical program at Osaka, Japan this year. **Example topics of interest includes but is not limited to the following:**

1. Big Data Science and Foundations

- Novel Theoretical Models for Big Data
- New Computational Models for Big Data
- Data and Information Quality for Big Data
- New Data Standards

2. Big Data Infrastructure

- Cloud/Grid/Stream Computing for Big Data
- High Performance/Parallel Computing Platforms for Big Data
- Autonomic Computing and Cyber-infrastructure, System Architectures, Design and Deployment
- Energy-efficient Computing for Big Data
- Programming Models and Environments for Cluster, Cloud, and Grid Computing to Support Big Data
- Software Techniques and Architectures in Cloud/Grid/Stream Computing
- Big Data Open Platforms
- New Programming Models for Big Data beyond Hadoop/MapReduce, STORM
- Software Systems to Support Big Data Computing

3. Big Data Management

- Search and Mining of variety of data including scientific and engineering, social, sensor/IoT/IoE, and multimedia data
- Algorithms and Systems for Big Data Search
- Distributed, and Peer-to-peer Search
- Big Data Search Architectures, Scalability and Efficiency

Source: bigdataieee.org/BigData2022/CallPapers.html

CONFERENCE AND SEMINAR

- **Call for papers:** the 16 seminar topics
- **Program committee:** the mentors
- **Peer review:** you review each other's papers
- **Conference:** the block seminar

Organization

Seminar Rules: Grading and Paper Requirements

~50%
Written paper

~50%
Presentation

One single grade for the seminar. The 50% / 50% weighting is **approximate**, not an exact calculation.

- Both components must be passed with at least grade **4.0**
- The grade can be **negatively** and **positively** affected by the **submitted reviews**, **template compliance** and **participation** in lecture

PAPER REQUIREMENTS

- LaTeX template: **IEEE**
- Length: at least **6** and at most **8** pages

Failure to meet either criterion leads to the immediate deduction of one grade level for the paper.

Organization

Seminar Rules: Dates, Reviews, Block Seminar and Participation

LECTURE AND SEMINAR DATES (MANDATORY)

Nov 3, 2026
Tuesday, 6 pm
Lecture: Scientific Work

Jan 13, 2027
Wednesday, 4 pm
Lecture: Presentation Training

Jan 29/30, 2027
Friday and Saturday
Block seminar

If you cannot make it to these dates, please **do not enroll** in the seminar. Attendance is **mandatory**. No exception.

Peer review

- Participation in the review process is **mandatory** for passing the seminar

Block seminar

- Each group gets a **15-minute slot**: at most 10 minutes of presentation, then open Q&A



- Attendance at the complete event is **mandatory**

Binding participation

- Once participation is confirmed (topic chosen and agreed with the mentor), you **cannot leave the seminar**; leaving counts as failed (**5.0**)
- A no-show at the block seminar also counts as failed

Motivation

- What is the greater context of the problem?
- Why do we want to find an answer?

Background and related work

- Explain the necessary background
- Which other solutions to this or similar questions exist?
- Why are they insufficient?

Placement of own work

- Demarcation: which aspects are addressed, which are not?
- How does the topic relate to the conference theme?
- Scientific question and sub-questions

Method and results

- Which scientific approach solves the problem, and why is it suitable?
- What results did the method yield, and what is the answer to the question?
- Further discussion: which aspects remain open?

Craft

- Precise language with abbreviations and technical terms explained
- Adequate, tidy graphics (labeled axes, size, colors)
- Adherence to the template
- No AI style, spelling, clear wording, pleasant reading flow

Formal requirements

- IEEE LaTeX template
- max. 8 pages

Source: S. Grundner-Culemann, Grading criteria for seminars, MNM-Team.

Organization

Grading Criteria: Presentation

Content

- Clear golden thread and clear message
- Presentation tailored to the audience
- Essential content of your work presented; detail balanced with abstraction

Craft

- Adherence to the template, slide numbering, appealing layout
- Slides support the talk but do not replace it (good visualizations, little text)
- Appropriate storytelling; useful information on the last slide

Speaking

- Professional expertise; speaking mostly without notes
- Suitable mix of prepared wording and free speaking
- Answers to questions show essential understanding of the research matter

Skill

- Clear pronunciation, suitable volume and speed, deliberate pauses
- Suitable gestures and posture; basic level of English
- Adherence to the time limit; brief yet informative answers

Source: S. Grundner-Culemann, Grading criteria for seminars, MNM-Team.

Motivation

Three Drivers of AI Progress: Algorithms, Data and Compute

Algorithms

- Transformer and self-attention (2017)
- Mixture-of-Experts
- Reinforcement learning

Data

- Open models pre-trained on 25 to 36T tokens (Nemotron 3, GLM-5, Qwen3)
- Public human text may be exhausted between 2026 and 2032
- RL environments with verifiable rewards (DeepSeek-V3.2: 1,800+ synthetic agent environments)
- Synthetic, simulation and robot data for physical AI

Compute

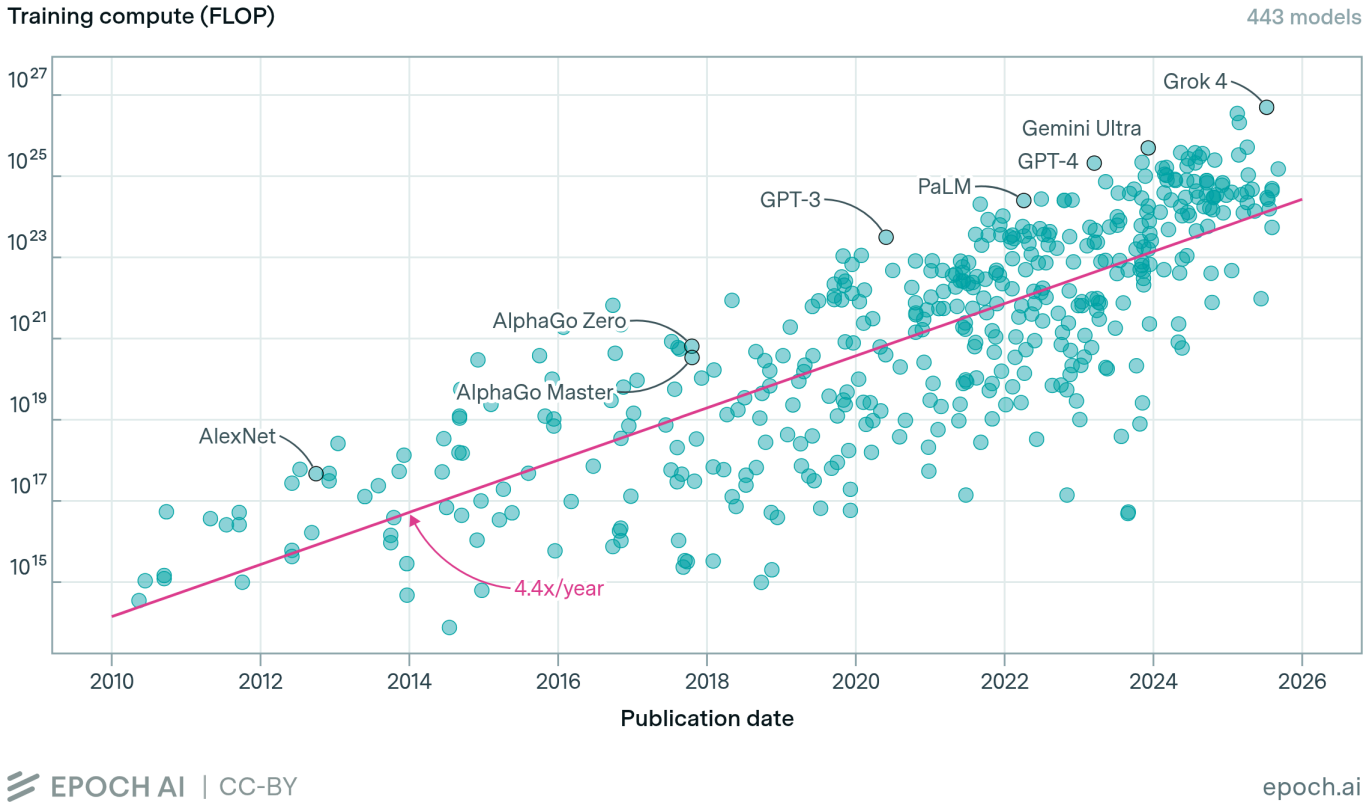
- Frontier training compute grows 4 to 5x per year
- Clusters with 100,000+ GPUs (e.g., xAI Colossus, 2024)
- Inference and test-time compute become a major cost factor
- Specialized accelerators: GPUs, TPUs, wafer-scale, ASICs

Hernandez and Brown (2020), Measuring the Algorithmic Efficiency of Neural Networks; NVIDIA (2025), Nemotron 3 Nano; GLM-5 Team (2026), GLM-5: from Vibe Coding to Agentic Engineering; Qwen Team (2025), Qwen3 Technical Report; DeepSeek-AI (2025), DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models; Schuhmann et al. (2022), LAION-5B; Villalobos et al. (2024), Will We Run Out of Data?; Epoch AI (2024), Training Compute of Frontier AI Models Grows by 4 to 5x per Year.

Motivation

Training Compute of Notable AI Models

Training compute of notable models



Source: Epoch AI (2024), Compute Trends Across Three Eras of Machine Learning.

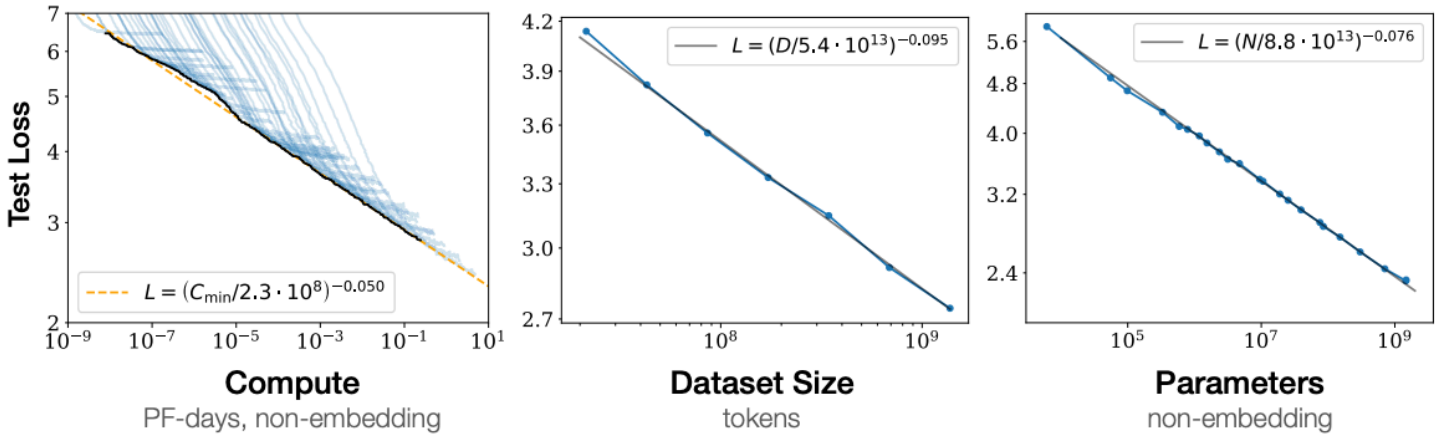
4 to 5x per year: growth of training compute for frontier models since 2010

~2.4x per year: growth of the cost of frontier training runs

Consequence: progress depends on efficient algorithms and large-scale distributed infrastructure **Topics 13 to 16**

Motivation

Scaling Laws and New Scaling Axes



Source: Kaplan et al. (2020), Scaling Laws for Neural Language Models.

Scaling law: model error falls predictably as we add compute, data or parameters.

Pre-training scaling

- Parameters, tokens and training compute
- Dense vs. sparse (MoE) models 1

Post-training scaling

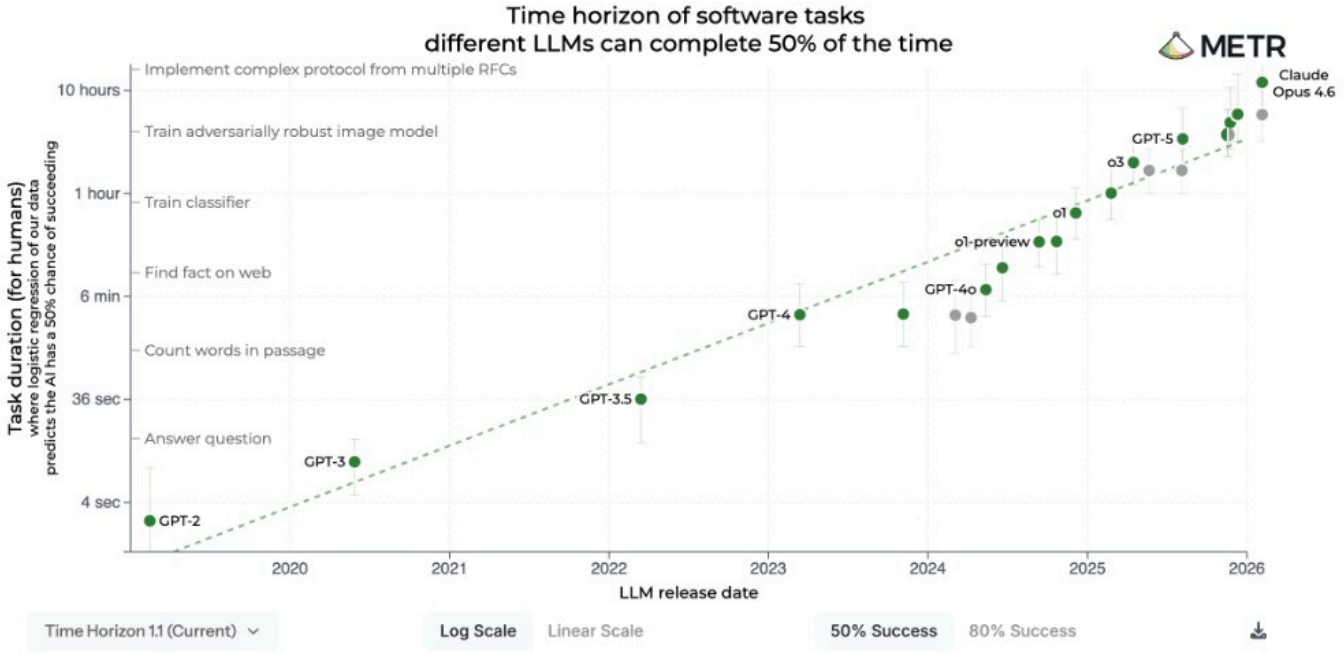
- RL with preference and verifiable rewards 6
- Distillation into smaller models 7

Test-time scaling

- Longer reasoning, search, self-verification 6
- Model routing 14

Motivation

AI Agents Complete Increasingly Long Tasks



Source: Kwa et al. (2025), Measuring AI Ability to Complete Long Tasks (METR).

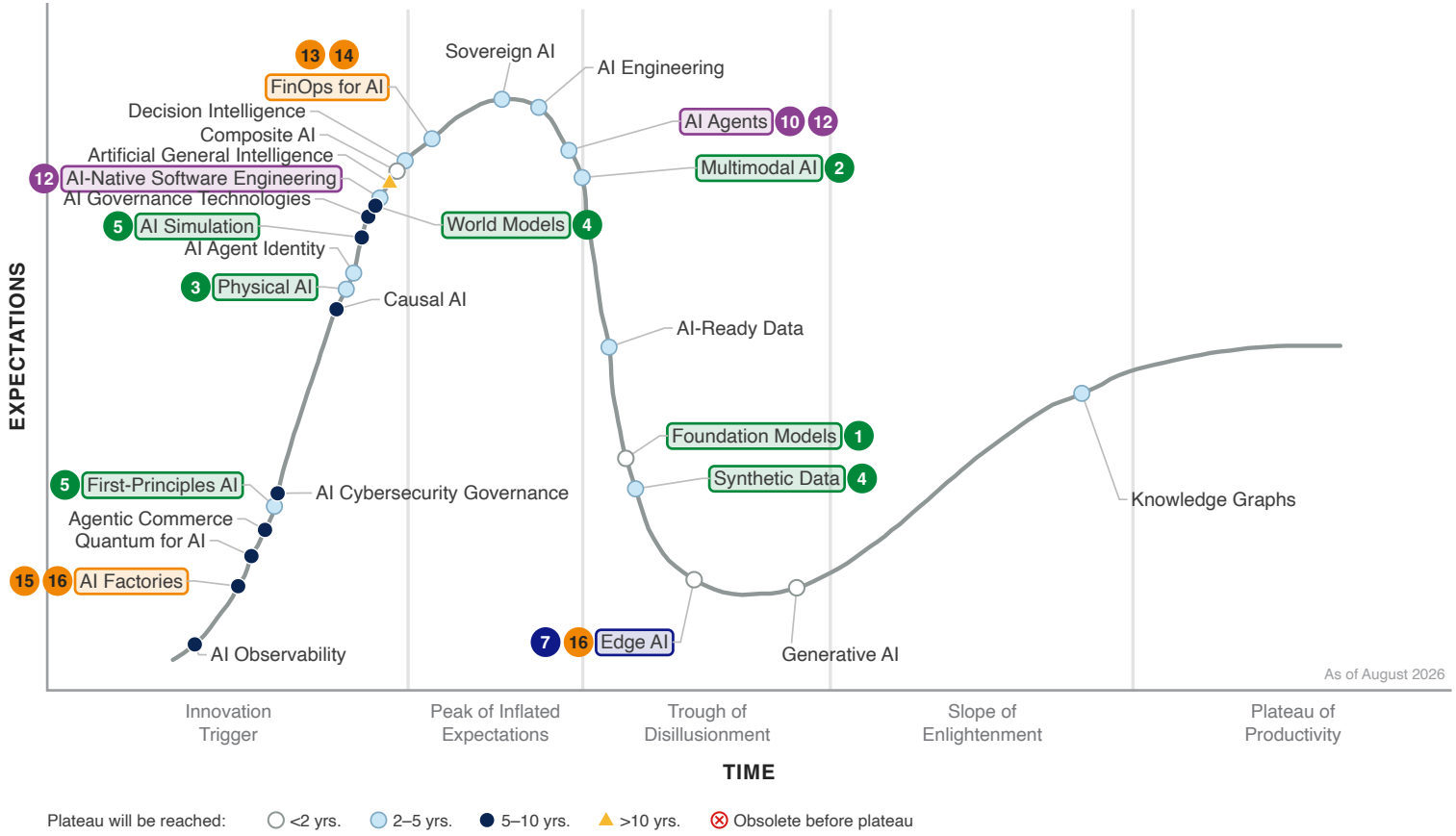
~7 months: doubling time of the task length that frontier models complete with 50% success

Long tasks require **tools, memory and robust harnesses** Topics 10, 12

Agents start to **improve algorithms and themselves** Topics 9, 11

Motivation

Where Are We? Gartner Hype Cycle for AI 2026



Source: Gartner (2026), Hype Cycle for Artificial Intelligence (redrawn). Highlights and numbers: related seminar topics.

- AI agents and multimodal AI have passed the **Peak of Inflated Expectations** 2, 10, 12
- Foundation models slide into the **Trough of Disillusionment** 1
- World models, physical AI and AI factories are still in the **Innovation Trigger** 3 to 5, 15, 16
- Seminar goal: separate **scientific substance** from hype by reading the primary literature critically

LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

Seminar Topics

Overview: 16 Topics in Four Areas

MNM TEAM

AREA 1
Foundation Models

1. LLM Architecture
2. Multi-Modal Foundation Models
3. Vision-Language-Action Models
4. World Models
5. Physics AI

AREA 2
Model Learning, Adaptation and Reasoning

6. Reasoning Models, RLHF and Test-Time Compute
7. Model Distillation and Parameter-Efficient Customization
8. Joint-Embedding Predictive Architectures (JEPA)
9. Recursive Self-Improvement

AREA 3
Applied and Agentic AI Systems

10. Agentic AI: Tools, Computer Use and Memory
11. AI Scientists and Automated Discovery
12. Agent Harnesses

AREA 4
AI Systems and Compute Infrastructure

13. Efficient LLM Inference and Serving
14. LLM Routing and Model Selection
15. Distributed Training and Communication
16. Specialized AI Accelerators

Area 1: Foundation Models

1. LLM Architecture

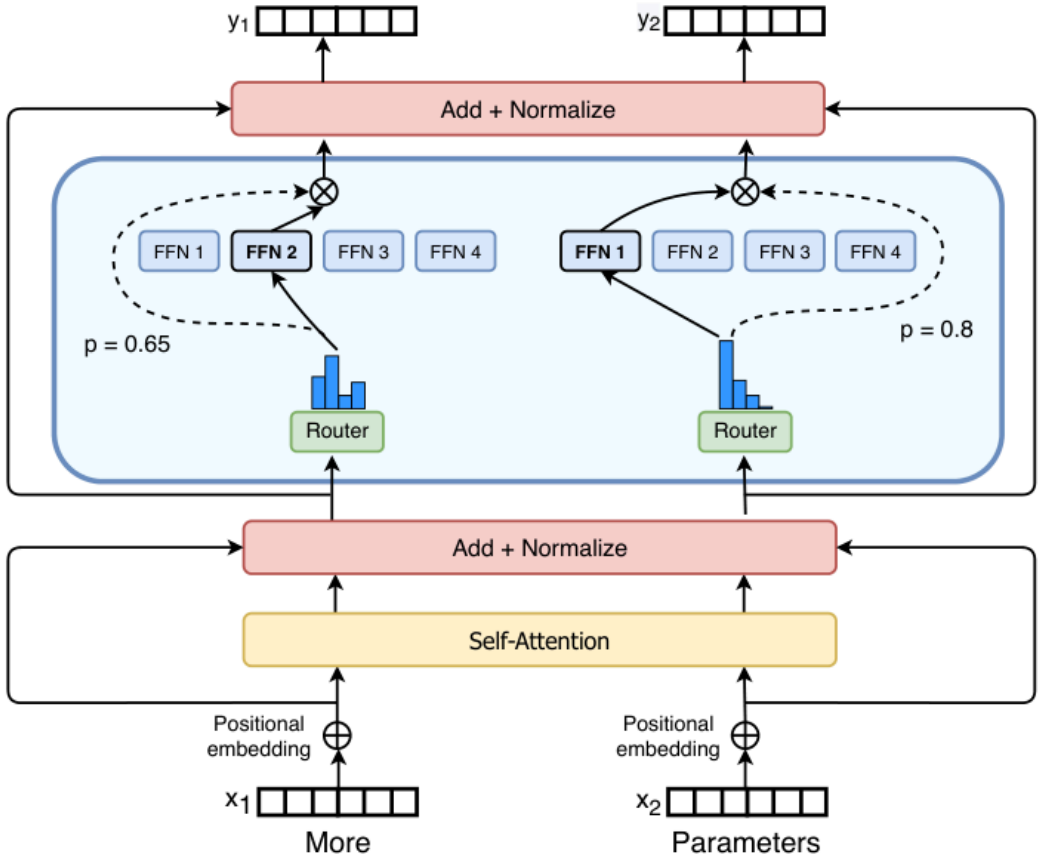
1 *Which architectural principles made modern large language models possible, and how do dense and sparse designs scale?*

KEY CONCEPTS

Transformer | Self-attention | Mixture-of-Experts

CORE PAPERS

- **Vaswani et al. (2017),** Attention Is All You Need.
- **Brown et al. (2020),** Language Models are Few-Shot Learners (GPT-3).
- **Fedus, Zoph and Shazeer (2021),** Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity.



Switch Transformer block: the dense feed-forward layer is replaced by sparse experts; a router sends each token to one expert. Source: Fedus et al. (2021).

Mentor: **Karl Fürlinger**

Area 1: Foundation Models

2. Multi-Modal Foundation Models

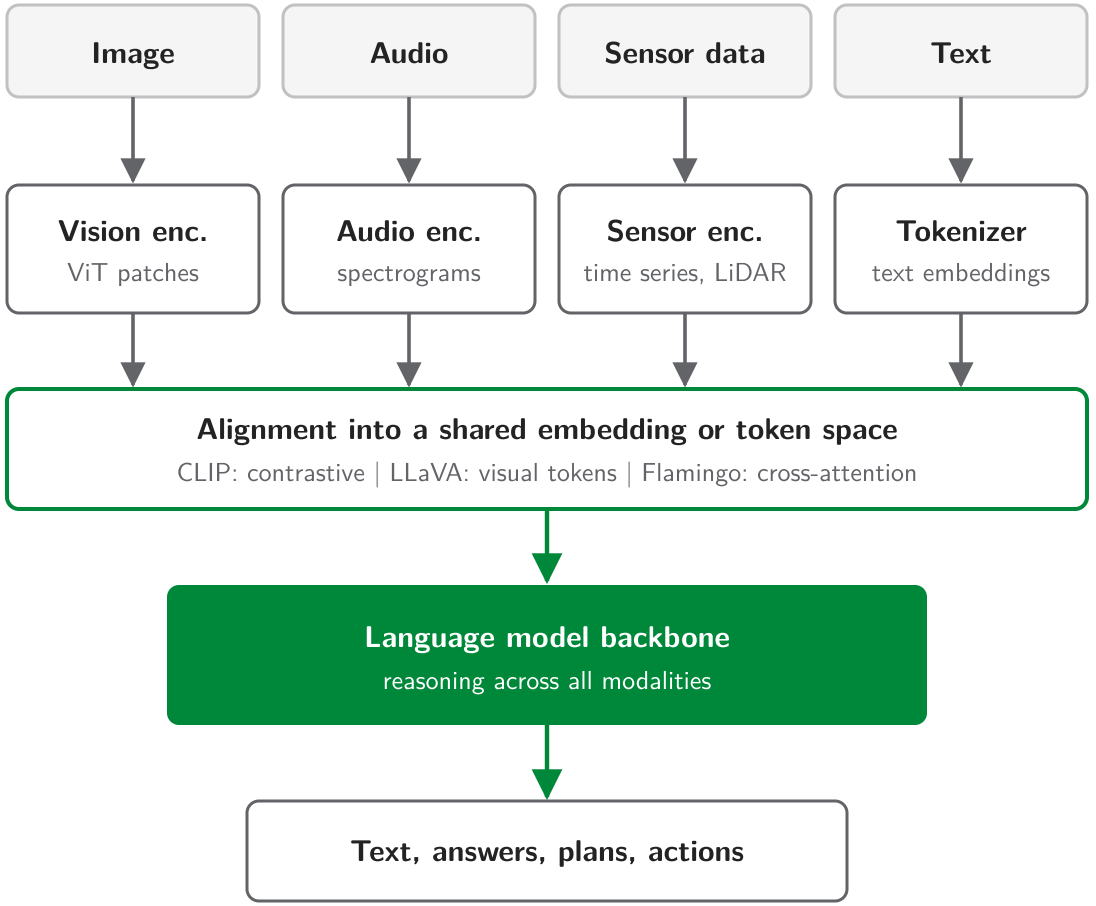
2 *How can foundation models connect heterogeneous modalities (e.g., text, audio, sensor data, vision) for perception, reasoning and interaction?*

KEY CONCEPTS

Contrastive learning | Cross-attention

CORE PAPERS

- **Radford et al. (2021),** Learning Transferable Visual Models From Natural Language Supervision (OpenAI CLIP).
- **Liu et al. (2023),** Visual Instruction Tuning (LLaVA).
- **Alayrac et al. (2022),** Flamingo: A Visual Language Model for Few-Shot Learning.



Schematic: common architecture pattern of multi-modal foundation models.

Mentor: **Fabian Dreer**

Area 1: Foundation Models

3. Vision-Language-Action Models

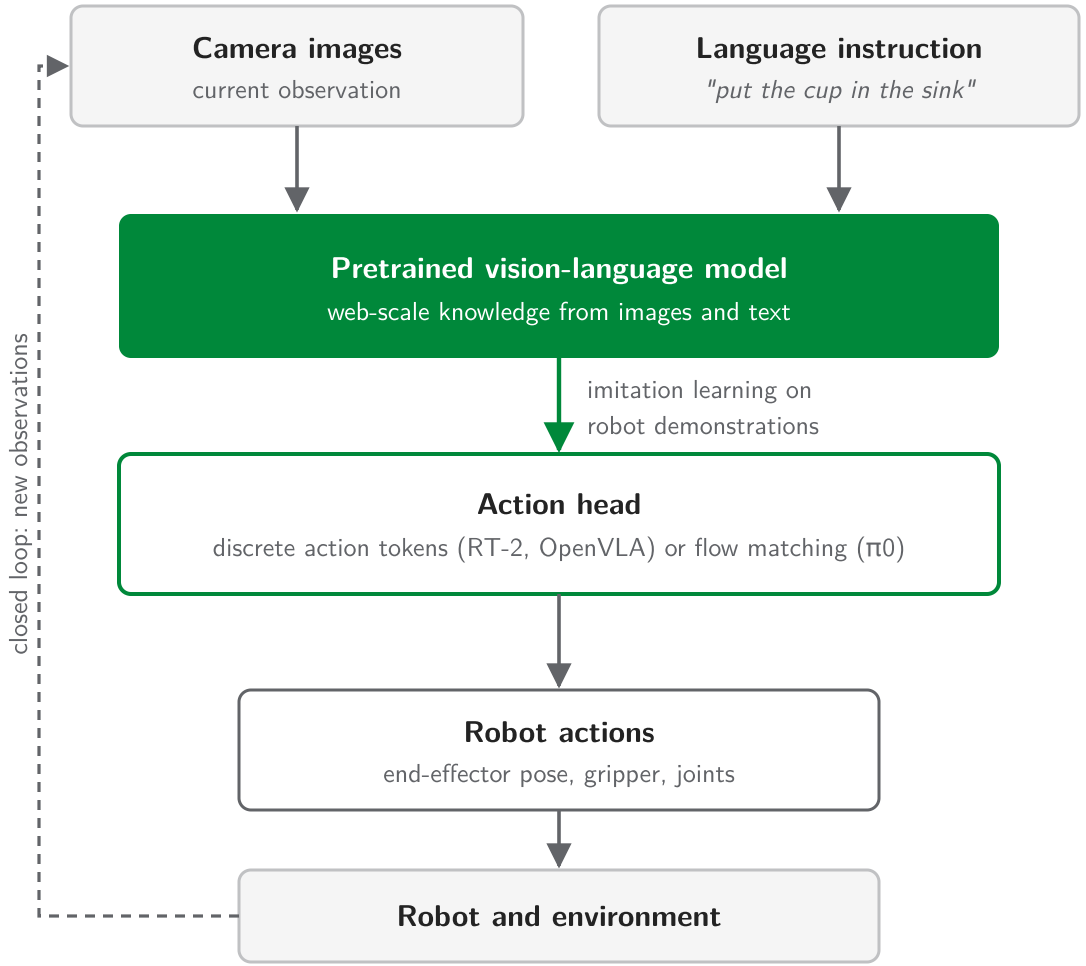
3 *How do vision-language models translate perception and language instructions into actions?*

KEY CONCEPTS

Actions tokens | Imitation learning

CORE PAPERS

- **Brohan et al. (2023),** RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control.
- **Kim et al. (2024),** OpenVLA: An Open-Source Vision-Language-Action Model.
- **Black et al. (2024),** π_0: A Vision-Language-Action Flow Model for General Robot Control.



Schematic: VLA models reuse a pretrained VLM and add an action output.

Mentor: **Daniel Diefenthaler**

Area 1: Foundation Models
4. World Models

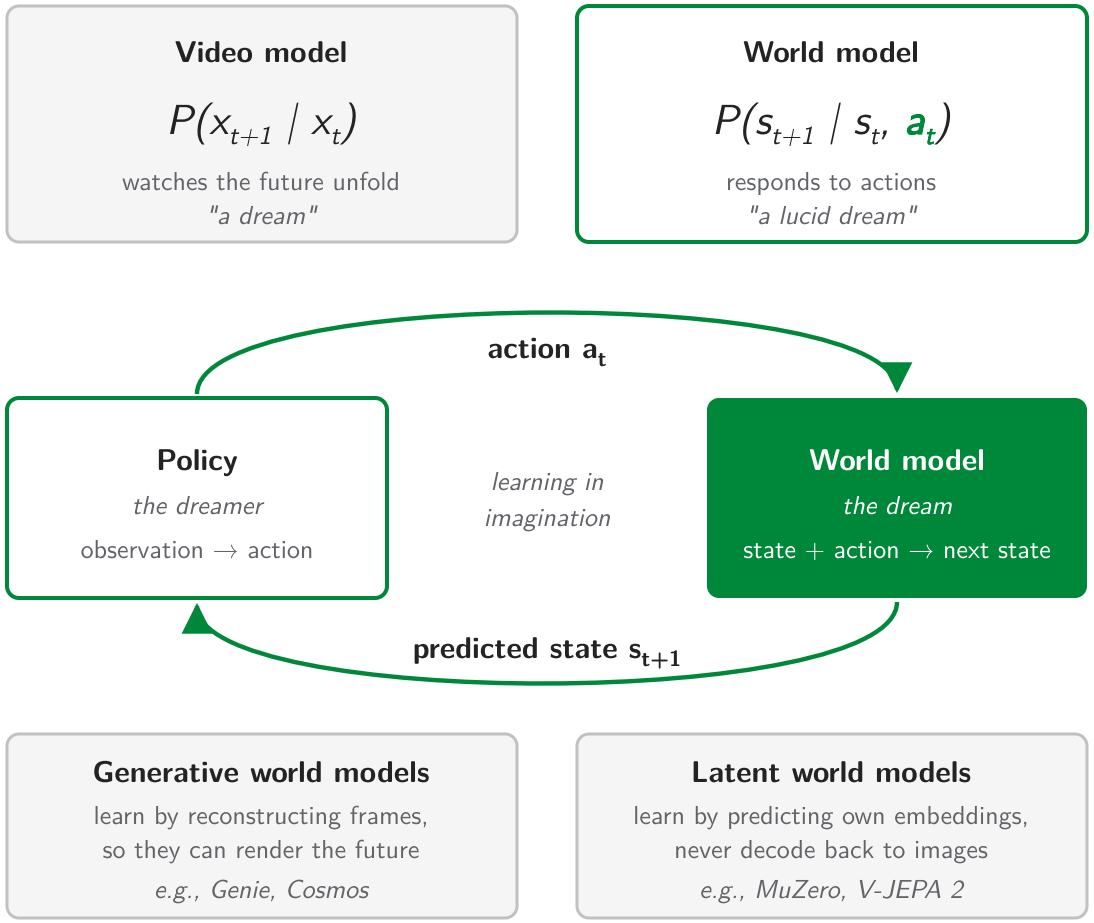
4 *How can models learn predictive representations of environments for simulation, planning and physical AI?*

KEY CONCEPTS

Action-conditioned video models | Learning in imagination

CORE PAPERS

- **Ha and Schmidhuber (2018),** World Models.
- **Bruce et al. (2024),** Genie: Generative Interactive Environments.
- **NVIDIA et al. (2025),** Cosmos World Foundation Model Platform for Physical AI.
- **McCormick and De Witte (2026),** World Models: Computing the Uncomputable, Not Boring [essay].



Schematic inspired by McCormick and De Witte (2026), World Models: Computing the Uncomputable, Not Boring.

Mentor: **Fabian Dreer**

Area 1: Foundation Models

5. Physics AI

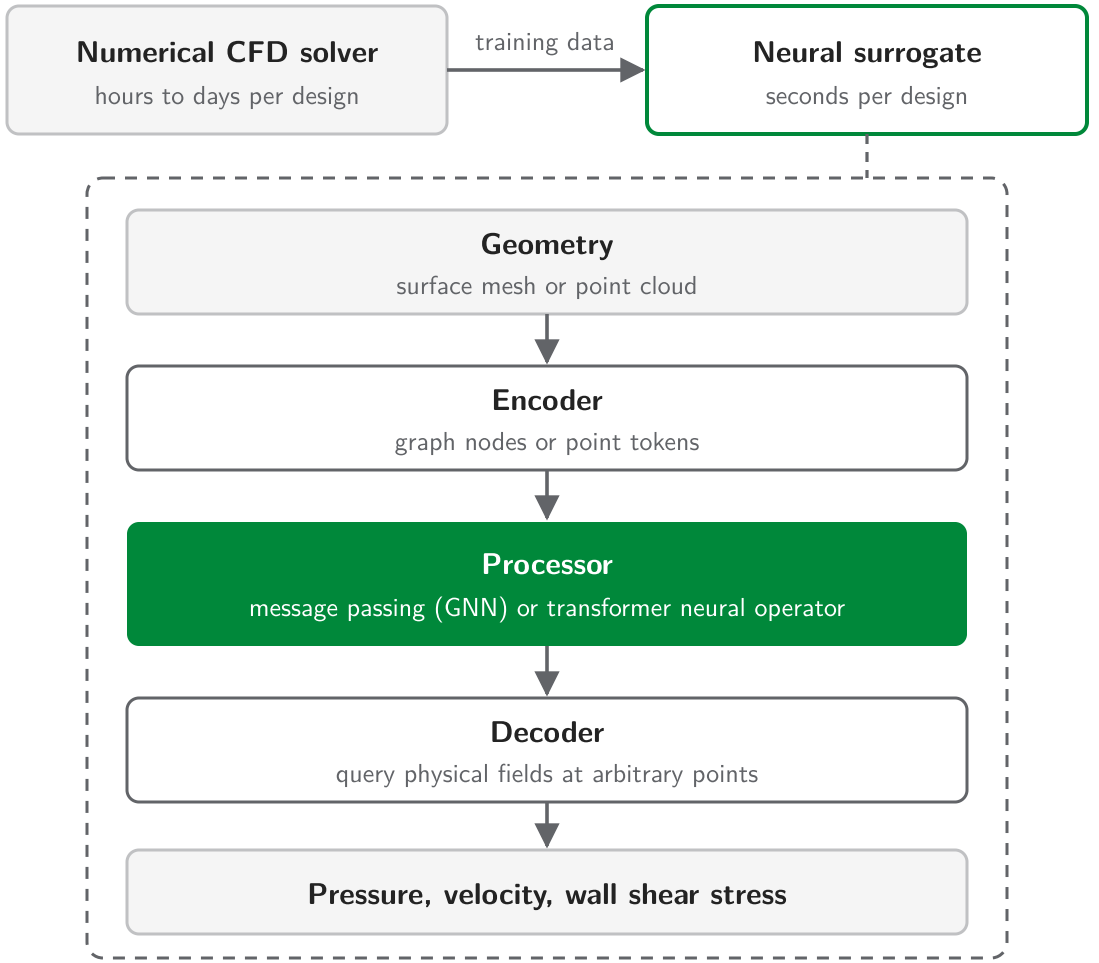
5 *How can graph networks and neural operators learn scalable surrogate models for complex physical systems?*

KEY CONCEPTS

MeshGraphNets | Neural operators | Physics transformer

CORE PAPERS

- **Pfaff et al. (2020),** Learning Mesh-Based Simulation with Graph Networks (MeshGraphNets).
- **Alkin et al. (2024),** Universal Physics Transformers: A Framework for Efficiently Scaling Neural Operators.
- **Alkin et al. (2025),** AB-UPT: Scaling Neural CFD Surrogates for High-Fidelity Automotive Aerodynamics Simulations.



Schematic: encoder-processor-decoder surrogates for physical simulation.

Mentor: **Sergej Breiter**

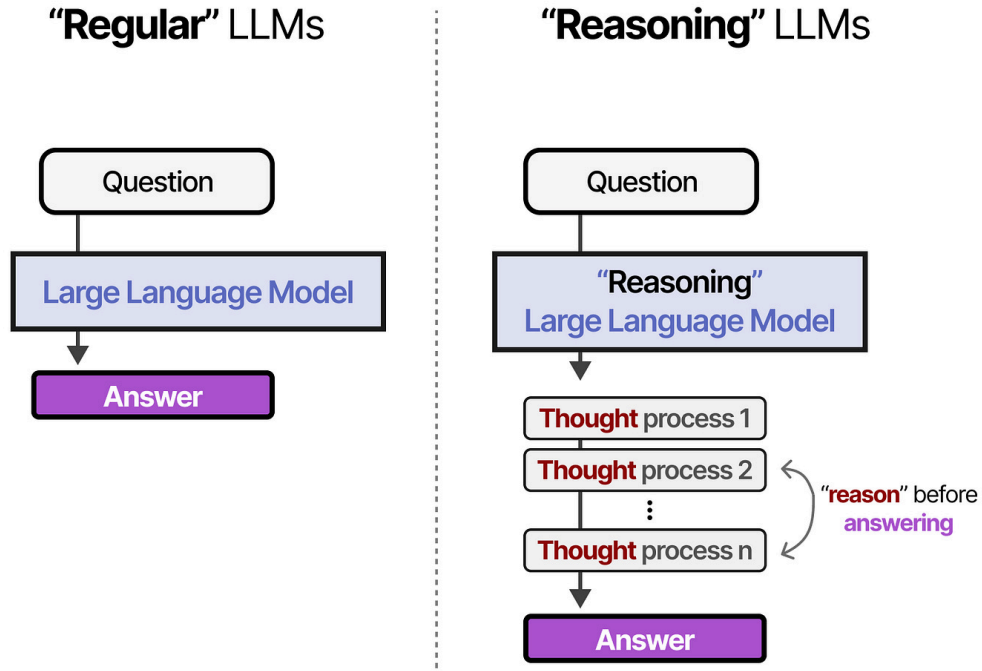
6 *How do preference learning, verifiable rewards and additional inference-time computation improve reasoning?*

KEY CONCEPTS

RL with human feedback | RL with verifiable rewards

CORE PAPERS

- **Lambert (2025 to 2026),** RLHF Book Course.
- **DeepSeek-AI (2025),** DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning.
- **Snell et al. (2024),** Scaling LLM Test-Time Compute Optimally Can Be More Effective than Scaling Model Parameters.



Reasoning models spend additional tokens on intermediate thoughts before answering. Source: Grootendorst (2025), A Visual Guide to Reasoning LLMs.

Mentor: **Kaiwei Lei**

7 *How do low-rank adaptation and distillation compare when specializing large language models?*

KEY CONCEPTS

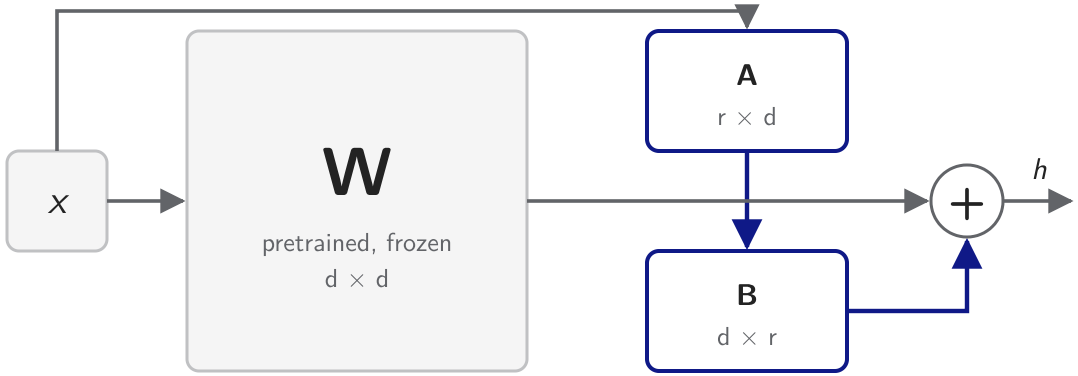
Low-rank adapters | Distillation

CORE PAPERS

- **Hu et al. (2021),** LoRA: Low-Rank Adaptation of Large Language Models.
- **Hsieh et al. (2023),** Distilling Step-by-Step! Outperforming Larger Language Models with Less Training Data and Smaller Model Sizes.
- **Zhao et al. (2026),** Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models.

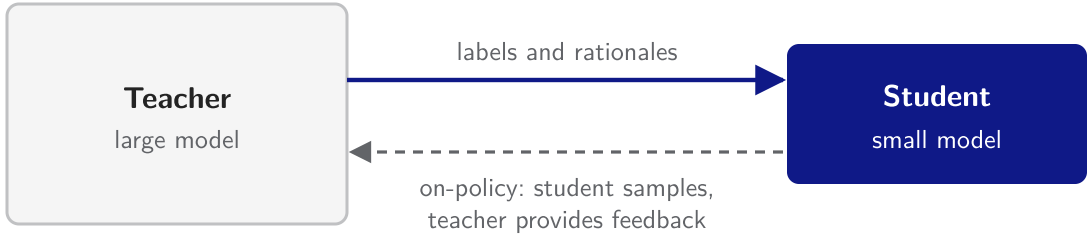
Mentor: **Kaiwei Lei**

LoRA: low-rank adaptation



$h = Wx + BAx$, rank r ≪ d: only A and B are trained

Distillation: teacher to student



Schematic: LoRA after Hu et al. (2021); distillation after Hsieh et al. (2023).

8. Joint-Embedding Predictive Architectures (JEPA)

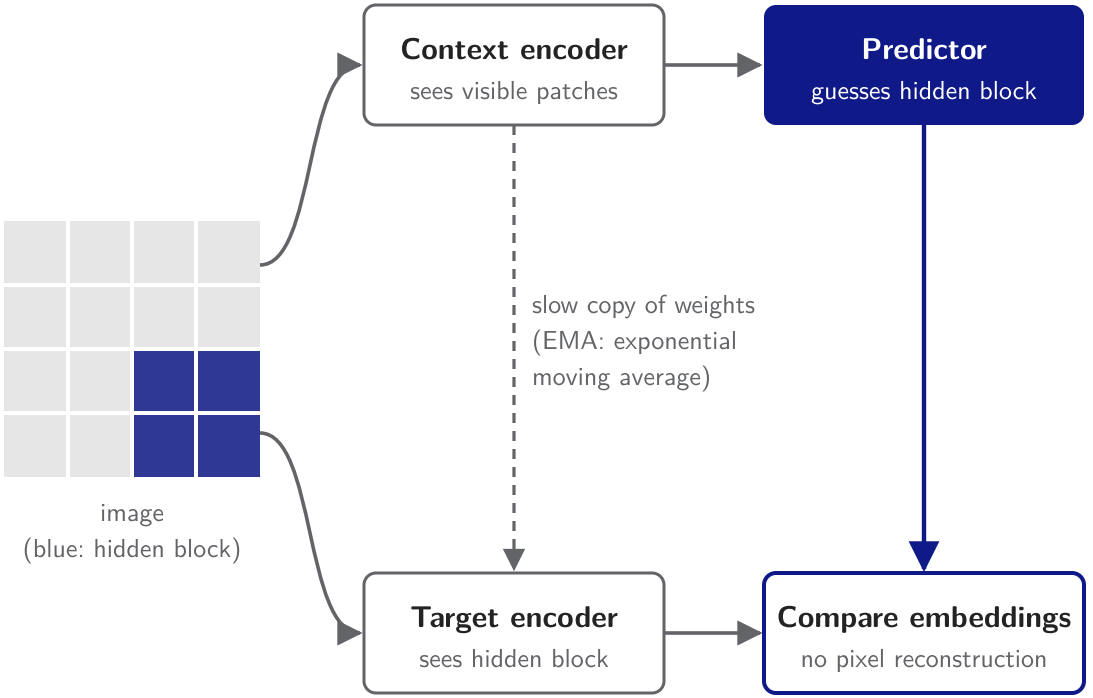
8 *Can representation-space prediction provide a scalable alternative to reconstructive and generative self-supervised learning?*

KEY CONCEPTS

Prediction in embedding space

CORE PAPERS

- **Assran et al. (2023),** Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA).
- **Assran et al. (2025),** V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning.
- **Balestriero and LeCun (2025),** LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics.



Generative self-supervised models reconstruct pixels; JEPA predicts embeddings.

Schematic after Assran et al. (2023), I-JEPA.

Mentor: **Sergej Breiter**

9. Recursive Self-Improvement

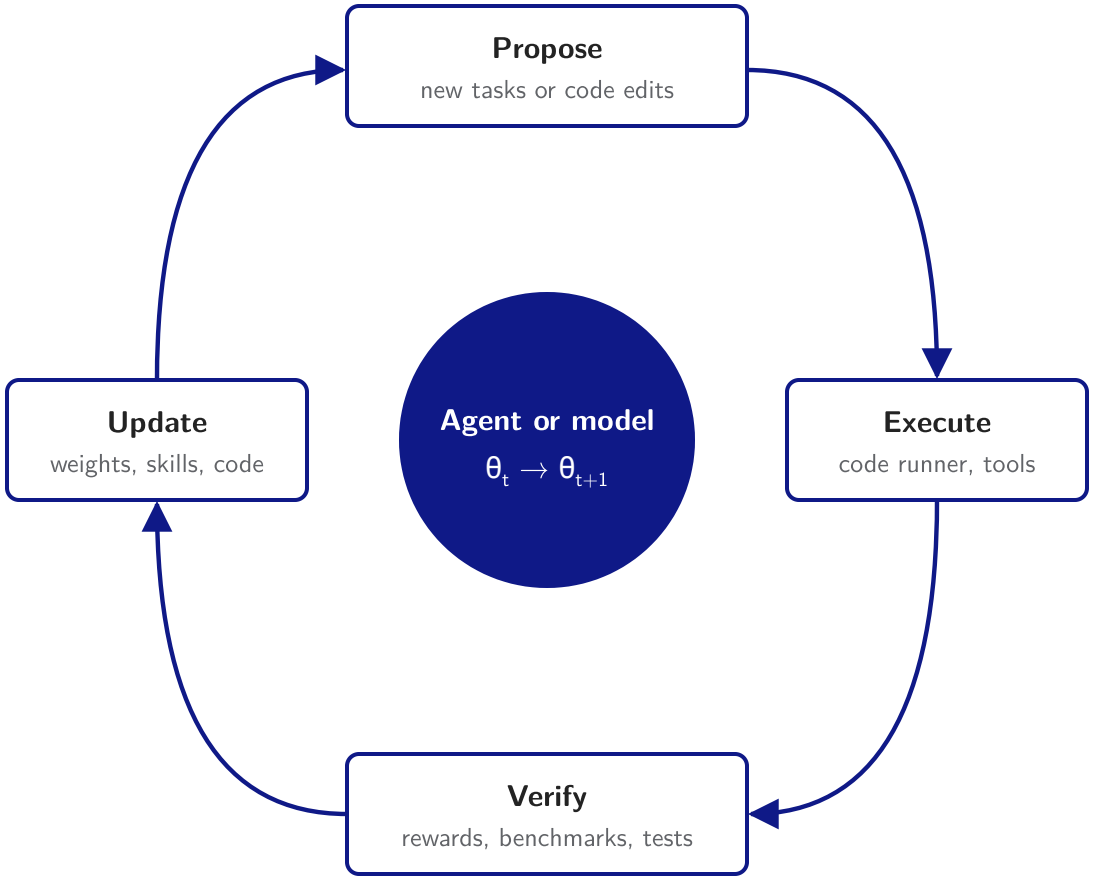
9 *When does self-play or self-modification produce genuine improvement rather than better search over fixed capabilities?*

KEY CONCEPTS

Self-play: proposer and solver | Self-modifying agents

CORE PAPERS

- **Zhao et al. (2025),** Absolute Zero: Reinforced Self-Play Reasoning with Zero Data.
- **Zhang et al. (2025),** Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents.
- **Wang et al. (2026),** MetaSkill-Evolve: Recursive Self-Improvement of LLM Agents via Two-Timescale Meta-Skill Evolution.



Genuine improvement or better search over fixed capabilities?

Schematic: common loop of self-play and self-modifying agents.

Mentor: **Andre Luckow**

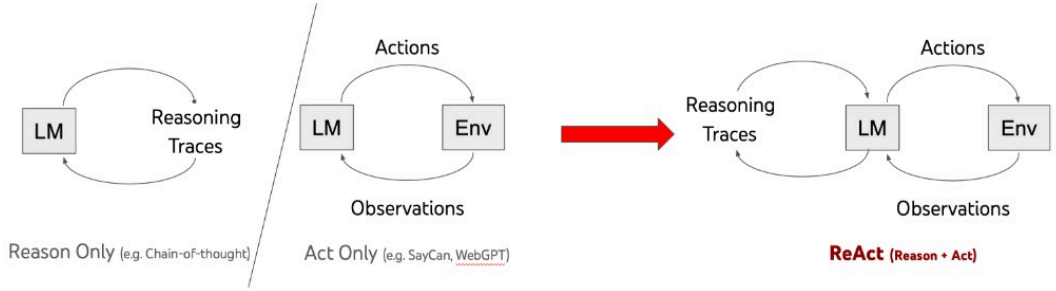
10 *How can agents plan, use tools, interact with computers and maintain useful state over long-running tasks?*

KEY CONCEPTS

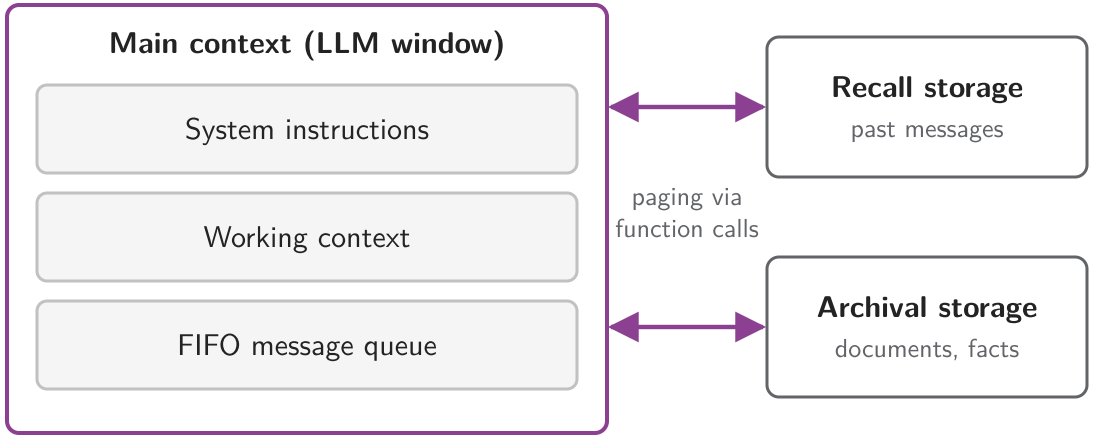
Agentic Architecture | Reason-act loop

CORE PAPERS

- **Yao et al. (2022),** ReAct: Synergizing Reasoning and Acting in Language Models.
- **Packer et al. (2023),** MemGPT: Towards LLMs as Operating Systems.
- **Zhang et al. (2025),** Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models.



ReAct interleaves reasoning traces and actions. Source: Yao et al. (2022).



Schematic: MemGPT memory hierarchy after Packer et al. (2023).

Mentor: **Daniel Diefenthaler**

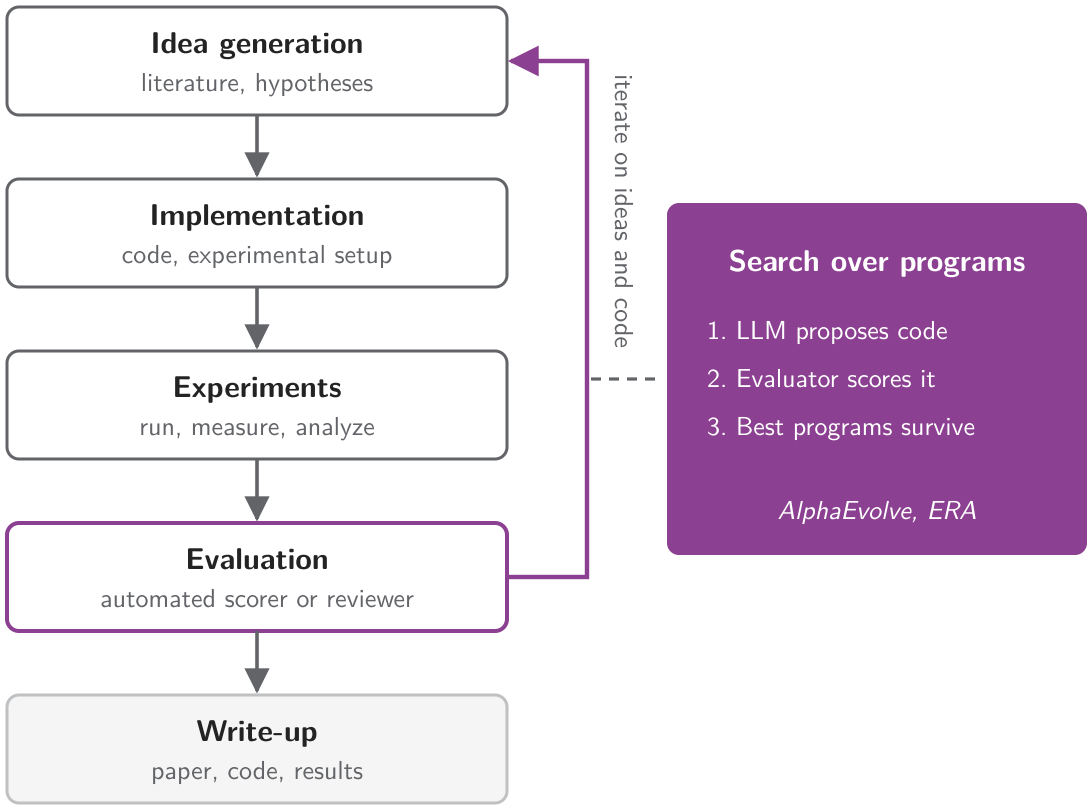
11 *How can AI systems formulate research ideas, run experiments, improve algorithms and produce scientifically useful discoveries?*

KEY CONCEPTS

Automated experimentation | Evolutionary algorithms

CORE PAPERS

- **Lu et al. (2024),** The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery.
- **Novikov et al. (2025),** AlphaEvolve: A Coding Agent for Scientific and Algorithmic Discovery.
- **Aygün et al. (2026),** An AI System to Help Scientists Write Expert-Level Empirical Software (ERA).



The AI Scientist automates the full research pipeline; AlphaEvolve and ERA focus on scored search over code.

Schematic: automated discovery loop.

Mentor: **Andre Luckow**

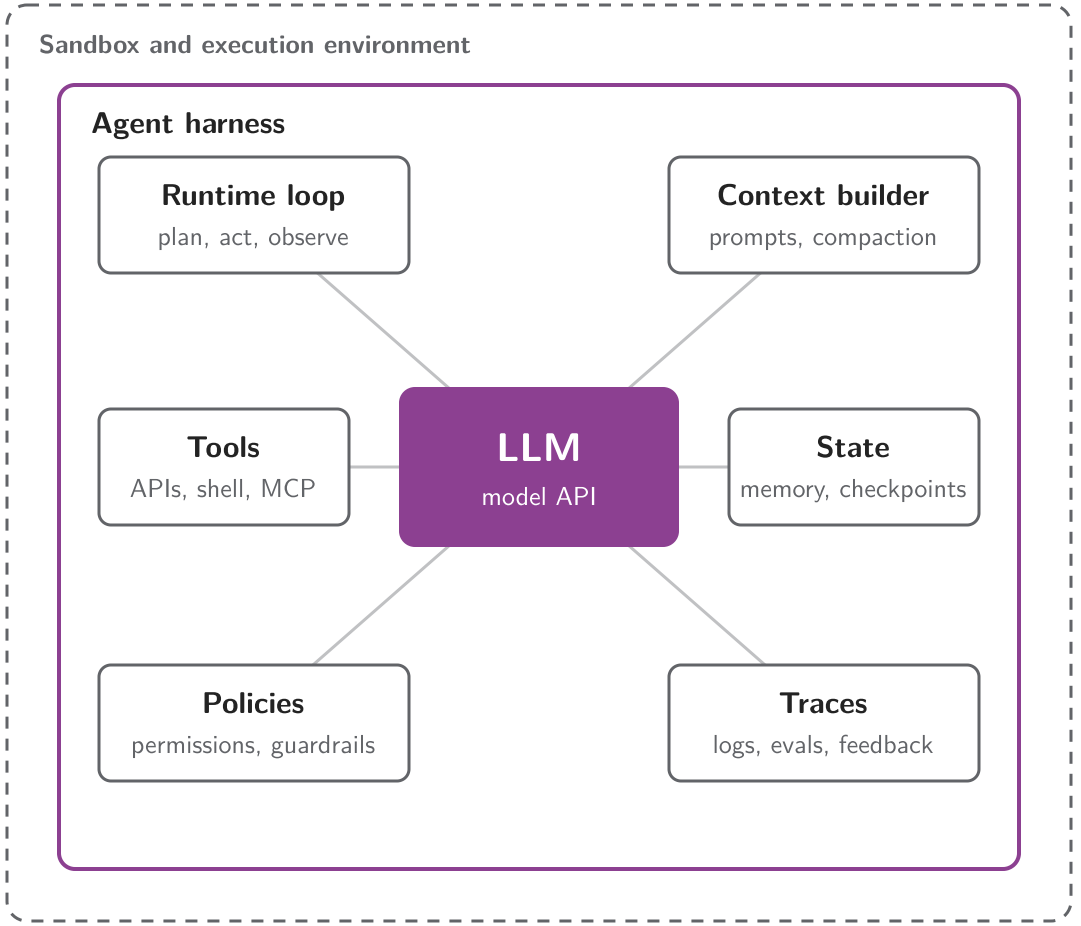
12 *How to architect and manage agentic runtimes?*

KEY CONCEPTS

Agent harnesses | Tool Use | Agent Security

CORE PAPERS AND SYSTEMS

- **ExoHarness (2026),** ExoHarness: A Minimal Architecture for Durable and Evolvable Agents [system/specification].
- **Omnigent (2026),** Omnigent Harness Architecture [system documentation].
- **Weng (2026),** Harness Engineering for Self-Improvement.



Schematic: components of an agent harness around the model.

Mentor: **Andre Luckow**

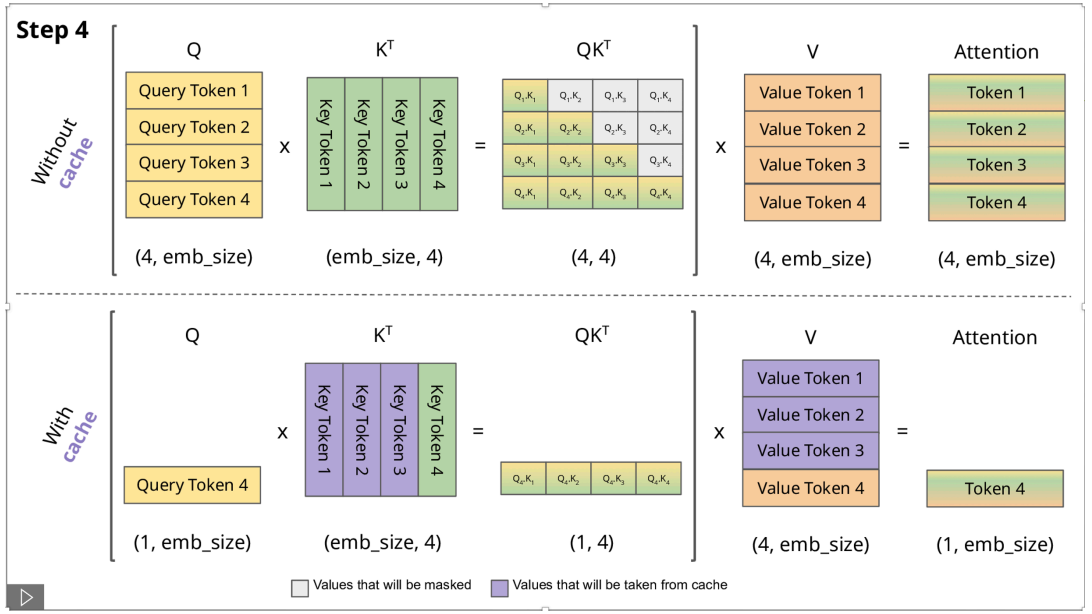
13 *How do modern serving frameworks manage KV caches, scheduling, structured generation, throughput and latency?*

KEY CONCEPTS

Attention Architectures | KV Cache | TTFT and throughput

CORE PAPERS

- **Kwon et al. (2023),** Efficient Memory Management for Large Language Model Serving with PagedAttention (vLLM).
- **Zheng et al. (2024),** SGLang: Efficient Execution of Structured Language Model Programs.
- **Majidi et al. (2026),** LLM Serving in the Wild: An Empirical Study of Frameworks, Methods, and System Designs.



KV caching: keys and values of previous tokens are stored and reused in each decoding step.
Source: Lages (2023), KV Caching Explained.

Mentor: **Josef Pichlmeier**

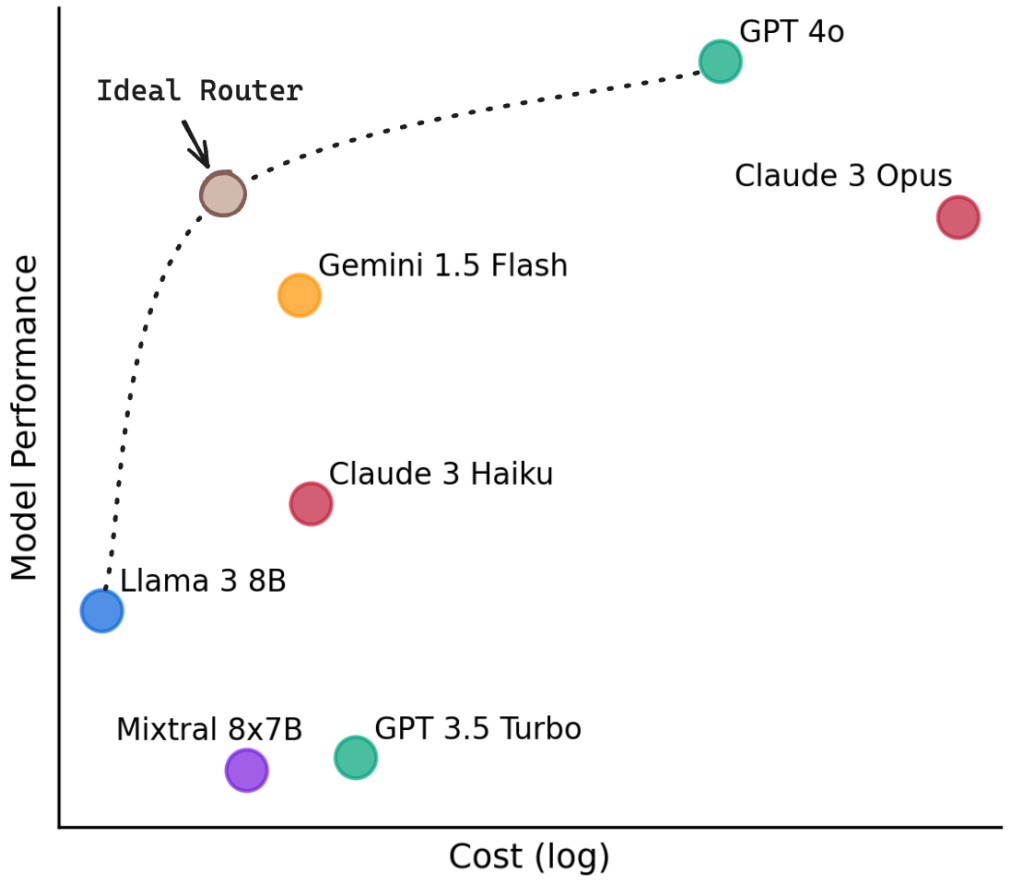
14 *How should a system select models and reasoning budgets under quality, cost and latency constraints?*

KEY CONCEPTS

Router Types | Pareto frontier

CORE PAPERS

- **Ong et al. (2024),** RouteLLM: Learning to Route LLMs with Preference Data.
- **vLLM Semantic Router Team (2026),** vLLM Semantic Router: Signal-Driven Decision Routing for Mixture-of-Modality Models.
- **Li et al. (2026),** LLMRouterBench: A Massive Benchmark and Unified Framework for LLM Routing [benchmark].



An ideal router dominates every single model on the cost-quality trade-off. Cf. Ong et al. (2024).

Mentor: **Josef Pichlmeier**

15 *How does data, tensor, pipeline and expert parallelism work?*

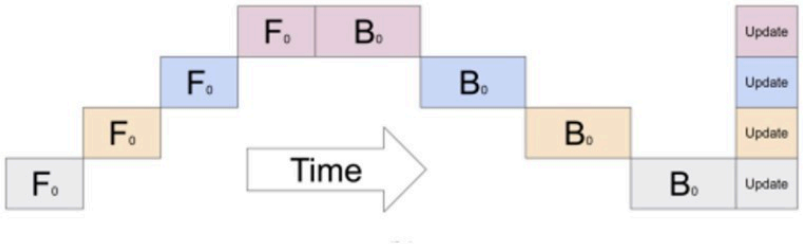
KEY CONCEPTS

Parallelism Types | All-reduce

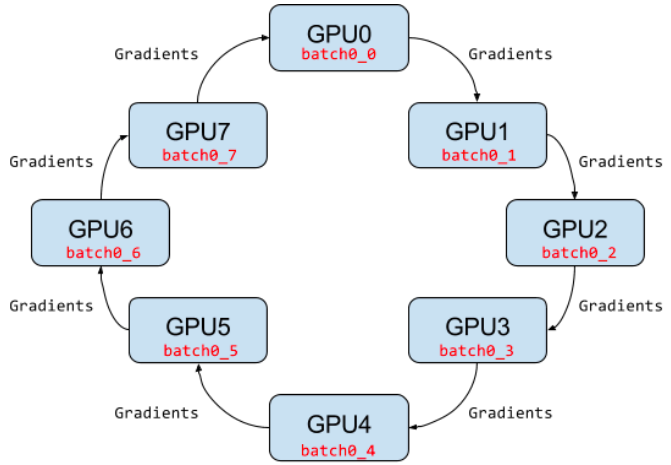
CORE PAPERS AND RESOURCES

- **Shoeybi et al. (2019),** Megatron-LM: Training Multi-Billion Parameter Language Models Using Model Parallelism.
- **PyTorch Team,** PyTorch Distributed Overview [framework documentation].
- **Hugging Face Nanotron Team (2025),** The Ultra-Scale Playbook: Training LLMs on GPU Clusters [course and empirical scaling guide].

Mentor: **Karl Fürlinger**



Pipeline parallelism: idle time ("bubble") between forward (F) and backward (B) passes. Source: Stanford CS336 (2025), after Huang et al. (2019).



All-reduce: gradients are synchronized across all workers. Source: Abadi et al. (2016).

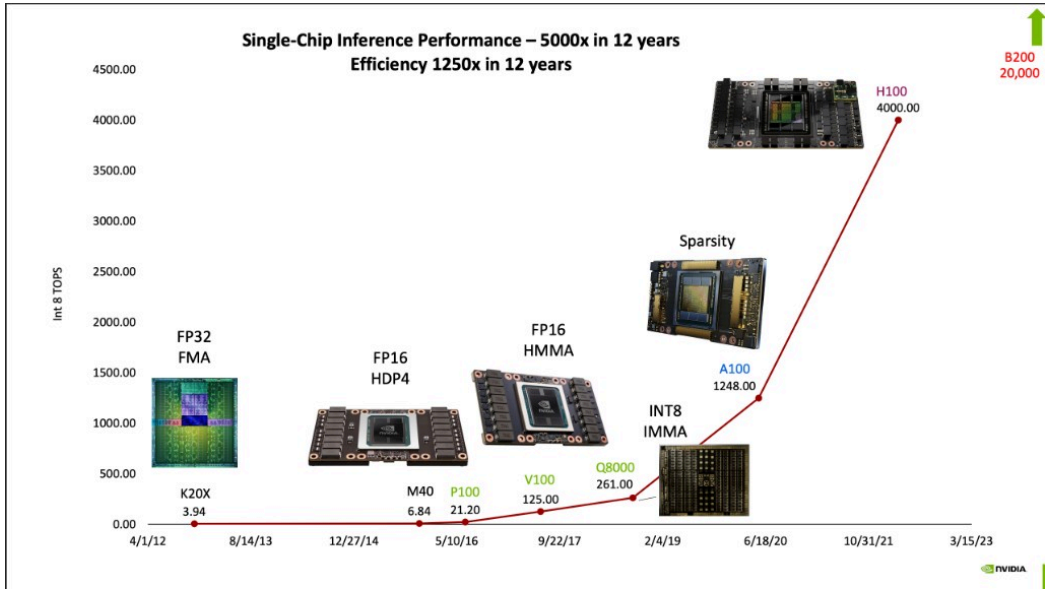
16 *How do accelerator designs trade off compute, memory, communication, programmability, energy and scale?*

KEY CONCEPTS

Tensor cores | Memory wall | Interconnects

CORE PAPERS

- **Jouppi et al. (2026),** Google's Training Supercomputers from TPU v2 to Ironwood: Architectural Stability, Scale, Resilience, Power Efficiency, and Sustainability Across Five Generations.
- **NVIDIA (2026),** NVIDIA Blackwell Architecture Technical Brief: Built for the Age of AI Reasoning.
- **Lie (2023),** Cerebras Architecture Deep Dive: First Look Inside the Hardware/Software Co-Design for Deep Learning.



Source: Dally (2024), DL Efficiency Through Technology.

GPU
many small matrix units (tensor cores) inside programmable cores, HBM, NVLink

TPU
few large systolic matrix grids that pass data cell to cell, HBM, optical network

Wafer-scale
one wafer-sized chip with large on-chip SRAM

Mentor: **Sergej Breiter**

Topic Selection

Topics and Mentors

ID	TOPIC	MENTOR
1	LLM Architecture	Karl Fürlinger
2	Multi-Modal Foundation Models	Fabian Dreer
3	Vision-Language-Action Models	Daniel Diefenthaler
4	World Models	Fabian Dreer
5	Physics AI	Sergej Breiter
6	Reasoning Models, RLHF and Test-Time Compute	Kaiwei Lei
7	Model Distillation and PEFT	Kaiwei Lei
8	Joint-Embedding Predictive Architectures	Sergej Breiter

ID	TOPIC	MENTOR
9	Recursive Self-Improvement	Andre Luckow
10	Agentic AI: Tools, Computer Use and Memory	Daniel Diefenthaler
11	AI Scientists and Automated Discovery	Andre Luckow
12	Agent Harnesses	Andre Luckow
13	Efficient LLM Inference and Serving	Josef Pichlmeier
14	LLM Routing and Model Selection	Josef Pichlmeier
15	Distributed Training and Communication	Karl Fürlinger
16	Specialized AI Accelerators	Sergej Breiter

Topic Selection

Next Steps and Working with Your Topic

TOPIC SELECTION

1. Review the topics and skim the three core papers
2. Submit your **four preferred topics by topic ID** (1 to 16) via Moodle
3. Conflicts are resolved by a random draw; assignments are announced promptly
4. Bachelor students (Proseminar) are grouped by the system
5. Meet your mentor to agree on scope and research question

WORKING WITH YOUR TOPIC

- Start from the **research question**, not from the papers
- Explain core idea, evidence and limitations of each paper
- Compare approaches systematically (e.g., feature matrix, benchmark results)
- Include recent follow-up work beyond the core papers
- Identify open problems and give your own assessment

> **The three core papers are the starting point, not the boundary of your topic.**